Assets workspace, consolidated

A tighter structure for ULTRON's Assets pages. Each fact gets one home; the other pages link to it instead of repeating it. Rules and limits live in Diagnosis, every AI / ML output lives in Analysis, and Prognosis is the brief. Ten machine tabs become eight (plus two kiln-only), with a new Maintenance tab. Wireframes show section order and the dominant visual per card.

What changes

The current build repeats the same content in up to five places. This is where each piece now lives.

Repeated today	Where it appears now	Single home in the proposal
Source · Updated · Inputs valid	Header meta and a KPI tile on every tab	One **machine bar** in the shell, under the tab row. Tabs show only tab-specific figures.
Hand-off buttons	Header of every tab (Health, Diagnosis, Analysis, Prognosis…)	Removed. The tab row is the navigation; sections link to a specific anchor only where a value needs context.
Machine drawing with pads	Overview › Machine and Sensors › Map	**Overview** only, with the validity / limit-state toggle moved onto it. Sensors keeps the register, no drawing.
Signal vs model table	Overview › Operating point, Health › Indicators, Diagnosis › Evidence, Trends › Signals, Sensors › Register	**Trends › Signals** is the full table. Overview keeps only process operating point; Diagnosis Evidence lists only sensors cited by a finding.
Health roll-up	Health › Breakdown, Overview group list, Diagnosis header	**Overview › Sub-assemblies** (health, Δ, points lost, leading signal). Health tab removed.
Health over time	Health › Trend, Prognosis › Degradation projection	**Trends › Health** for history; **Analysis › Remaining life** for the forward projection.
Rules and points lost	Diagnosis rule board, Health › Deductions, Thermal › Rules, Analysis › Rule trace	**Diagnosis › Rules**, with a *Points lost* column. Thermal rules are a filter on it.
Ranked faults / failure modes	Diagnosis signature match, Analysis › Diagnosis, Prognosis ›	**Analysis › Faults** owns the ML ranking and evidence matrix; **Prognosis › Risk** shows only probability × consequence. Diagnosis

	Failure risk table	keeps no model output.
Why (detection, explanation)	Analysis tab, half of it duplicating Diagnosis	**Analysis** is the single AI / ML tab and the longest page: one section per capability in the specs (health, expected behaviour, unknown anomaly, issue source & fault, degradation, future condition, failure risk, RUL, maintenance recommendation, explainability) plus model quality and feedback. **Diagnosis** is rules and limits only; **Prognosis** is a one-screen brief of capabilities 5–9 with links into Analysis.
Actions and operator feedback	Diagnosis › Actions, Analysis › Actions & feedback	**Maintenance › Actions**: one work list fed by rules (Diagnosis), AI recommendations (Analysis) and operators. **Analysis › Feedback** keeps the operator labels.
Event log	Diagnosis › Log, Trends › Events, Simulation › Event log, Analysis › Episodes	**Trends › Events**, filterable. Other pages link to it with a filter preset.
Model accuracy	Prognosis › Accuracy, Analysis › Model quality, AssetIQ › Accuracy	**Analysis › Model quality** for the machine; AssetIQ › Accuracy stays the fleet view.
Sensor validity	Sensors › Validity, Diagnosis validity strip, Health input quality, Rack channel quality, Thermal › Scanner	**Sensors › Validity**. Rack shows hardware state only; Thermal scanner coverage becomes one panel in Shell.
Maintenance (new)	Scattered: Diagnosis actions, Analysis recommendation, Prognosis window, Home work orders; no part life, shutdown or ROI tracking	New **Maintenance** tab from the MOM brief: actions, part life with ORANGE / YELLOW / GREEN, cycles, shutdowns and replacement timing, survey / maintenance records, ROI.
Unused modules	`prog-maintenance`, `prog-confidence`, `an-source`, `an-degradation`, `an-maintenance`, `an-xai`, `an-overview`	Deleted. Maintenance window and recommendations live on the Maintenance tab; confidence sits in Analysis › Remaining life.

One home per fact
If a number is shown on two tabs, one of them becomes a link.

Shell carries shared state
Health, lead finding, source and freshness render once in the machine bar, not per tab.

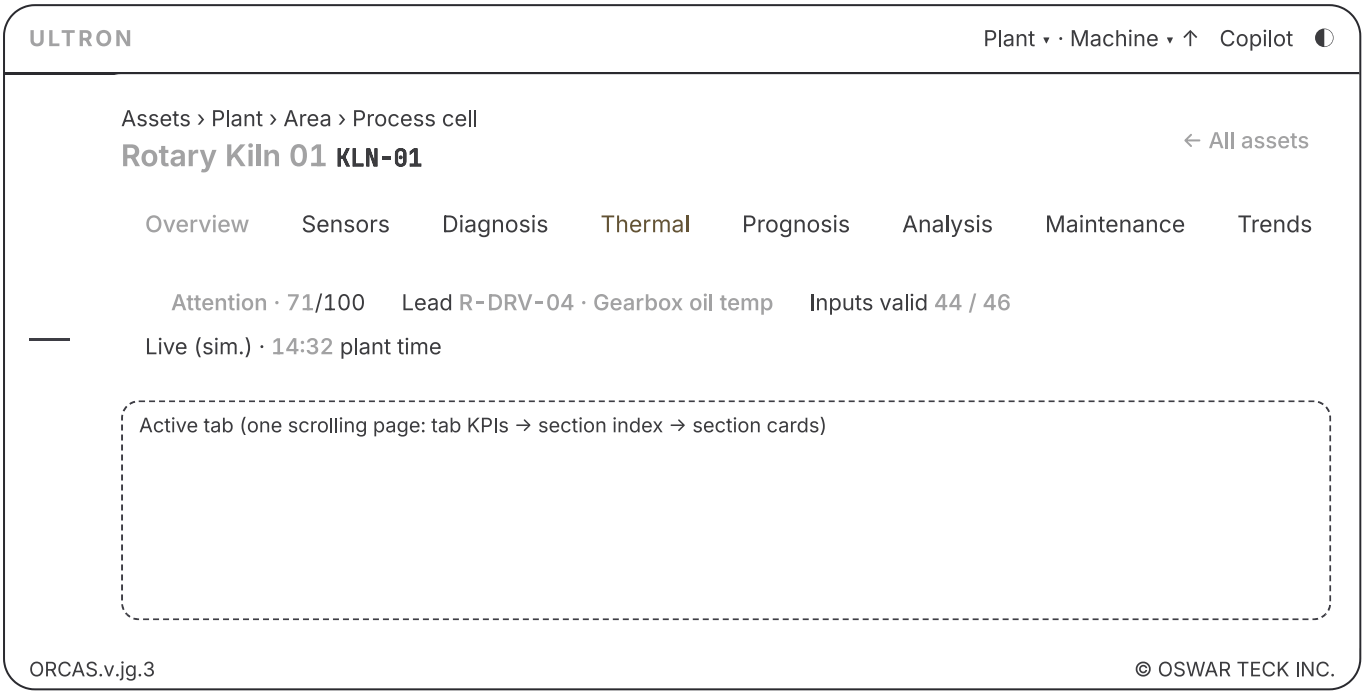
Four KPIs max per tab
Only figures that change the reading of that tab.

Sections hide when empty

Unchanged from today; it is what keeps non-kiln machines readable.

Machine shell `assets-view.tsx`

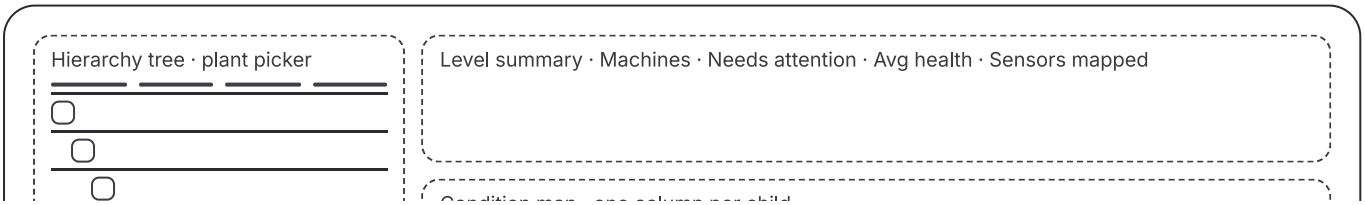
The frame every tab renders inside. The new machine bar replaces the per-tab header meta and the shared KPI tiles.

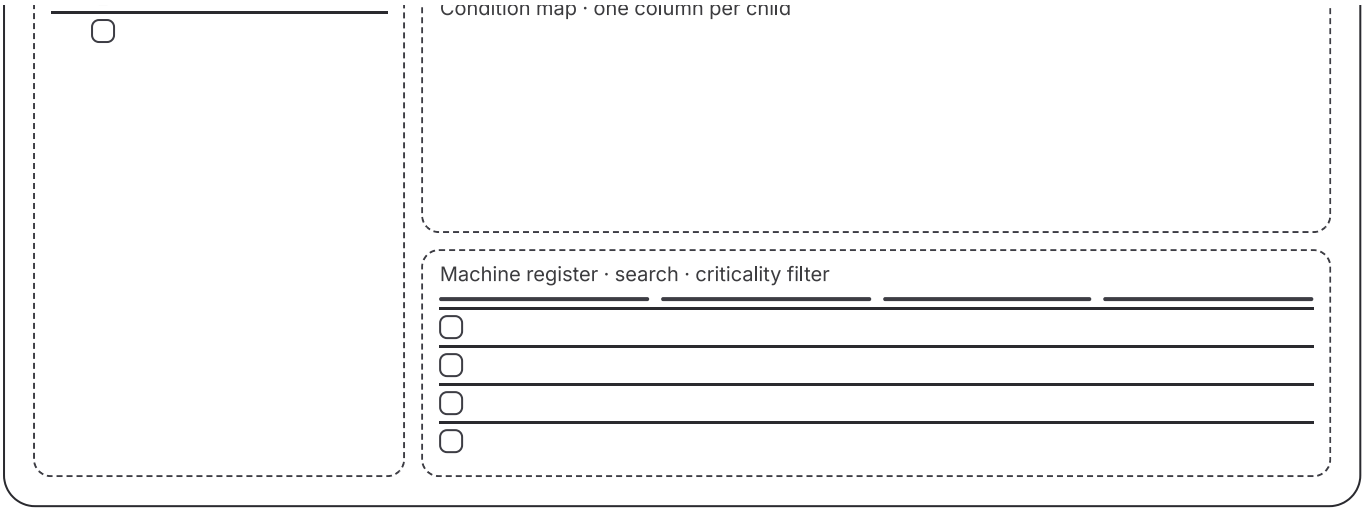


Header	Breadcrumbs, machine name + ID, *All assets*. Unchanged.
Tabs	Overview · Sensors · Diagnosis · Prognosis · Analysis · Maintenance · Trends · Rack. Diagnosis = deterministic rules, Prognosis = brief, Analysis = every AI capability in detail (the longest page), Maintenance = all work, parts, shutdowns and ROI. Kiln machines add Thermal (after Diagnosis) and Simulation (last). Health is gone.
Machine bar	New. Condition + health score, lead finding, inputs valid, source and plant time. One source: `liveMachineStatus()`. Clicking the lead finding opens Diagnosis on that rule.
Tab page	Tab-specific KPIs (max four) → sticky section index → one card per section. No header meta, no hand-off buttons.
Cross-links	Keep the sessionStorage hand-off (tag, anchor, rule ID). Links live inside sections, next to the value that needs context.

Directory `unchanged` `hierarchy/`

Question. What equipment do I have, where is it, and what state is each asset in?





Shown when No machine selected

Change None. Level summary, condition map and register do not overlap.

---

01 **Hierarchy tree**

- Plant → area → process cell → machine; health dot per node; drawer on phones

---

02 **Level summary**

- Four counts for the selected level

---

03 **Condition map**

- Machine tiles coloured by condition; click opens the machine

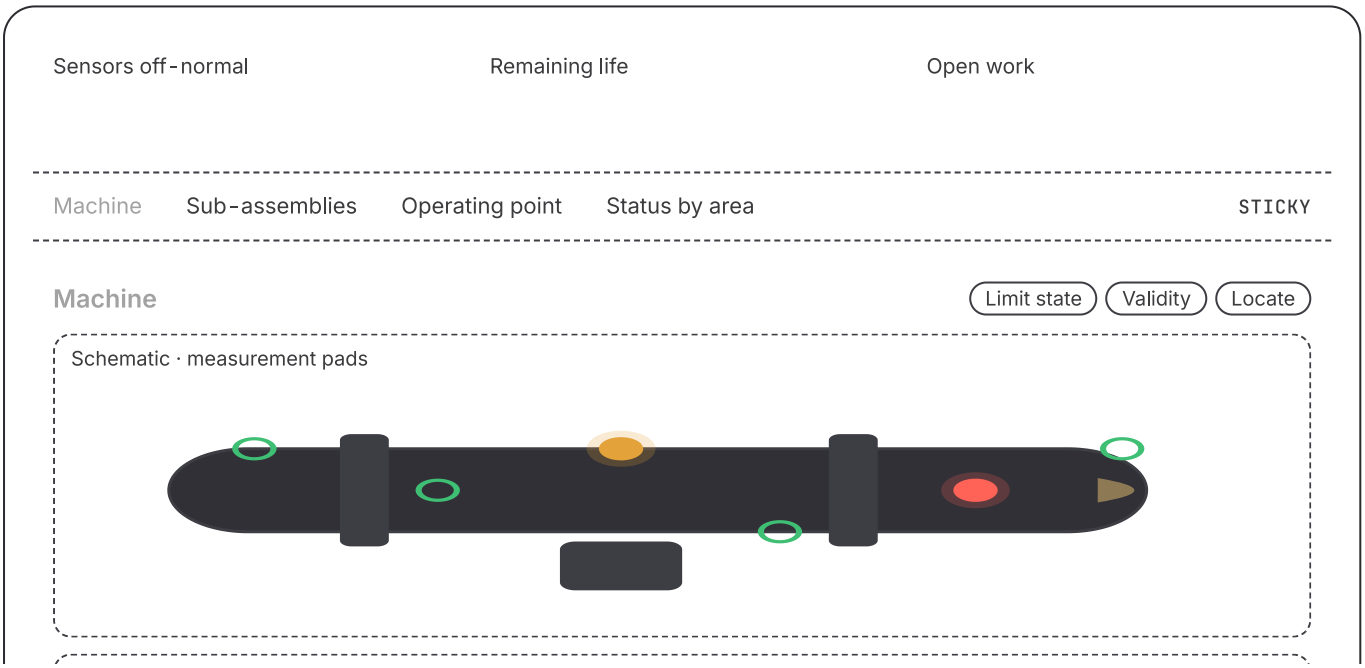
---

04 **Machine register**

`Columns: Machine · Location · Status · Health · Criticality · Sensors`

Overview `overview-page.tsx`

Question. Is it OK, and where on the machine is the problem?



Selected pad: P# · TAG · value · status · Trend / Diagnosis

P7 · TMP-04 312 °C Trends Diagnosis

Sub-assemblies

By system Lowest first

System → sub-assembly roll-up (was Health › Breakdown)

Operating point

Process parameters only

Status by area

One row per tab · value · state · →

Absorbs Health › Breakdown (roll-up table) and Sensors › Map toggle.

Drops Health index and inputs-valid KPIs (machine bar), the Health button.

---

01 KPI strip

- Sensors off-normal (alarm · warning)
- Remaining life of the governing component
- Open maintenance actions (→ Maintenance)

---

02 Machine

- Machine-kit drawing with TSE pads; toggle colours pads by limit state or data validity
- Hover a sub-assembly row to highlight it; click a pad for its callout with Trend / Diagnosis links

---

03 Sub-assemblies

Columns: Asset · Health · Δ · Lost · Leading signal · Findings · Status

- Grouped by system or flat, lowest first (D5 §9.4)
- Row click opens Diagnosis scoped to that sub-assembly

04 Operating point

Columns: Parameter · Value · Reference · Dev. · Normal · range · State

- Only D4 process signals (speed, feed, fuel, O_2…). Condition signals live in Trends › Signals

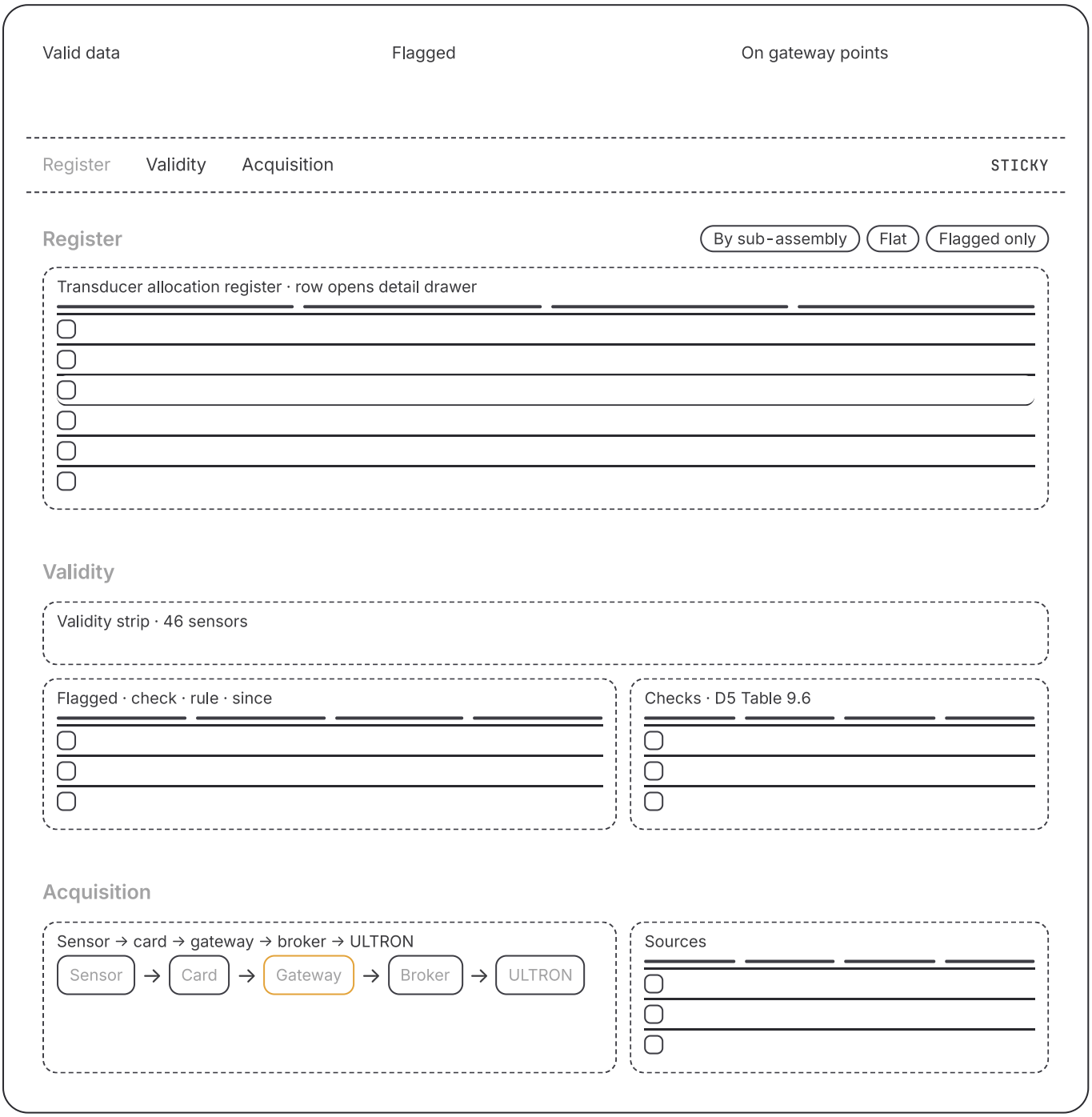
05 Status by area

Columns: Area · Value · Detail · State · →

- One-line digest of each tab, linking to it. Kept because it is the only cross-tab summary

Sensors `machine-sensor-mapping-view.tsx`

Question. Which sensors does this machine have, can I trust them, and where does each come from?



Drops — Map section (the drawing is on Overview). Limit-state filter (limits are a Trends concern).

Becomes the home of Sensor validity. Diagnosis, Rack and Thermal link here instead of showing their own validity views.

`01` **Register**

`Columns: Sensor ID · Measurement · Location · Range · Rack slot · reg. · Sampling · Data`

- Detail drawer: gauge, last sample, sparkline from MQTT history, links to Trend and Rack channel

---

`02` **Validity**

`Columns: Sensor · Data · Check · Rule · Since · Health input · Finding`

- Strip of every sensor; flagged table; the D5 Table 9.6 check definitions

---

`03` **Acquisition**

`Columns: Source · Kind · protocol · Gateway`

- Chain diagram warns when wired sensors have no gateway point

Diagnosis `diagnosis-view.tsx`

Question. What is wrong right now, according to the rules and limits, and what is the evidence?

Active alarms | Rules firing | Sensors in excursion | Vibration zone

Alarms Rules Evidence — `RULE SCOPE CHIP · STICKY`

Alarms

1 h | 8 h | 1 shift

Alarm performance

Alarm list · lifecycle · persistence counter

- [ ]
- [ ]
- [ ]
- [ ]

Rules

All | Process | Data quality | Thermal

21-rule board · lane · persist / clear · points lost · logic

- [ ]
- [ ]
- [ ]
- [ ]
- [ ]

Evidence

Sensors cited by a firing rule · value vs limits · residual 1 h

Vibration severity · ISO 10816-3 / 20816-3 zones

Scope	Deterministic only: the D5 rule library, alarm limits (Table 9.3), sensor validation and ISO vibration zones. Nothing here is a model output.
Moved to Analysis	Failure-mode signature ranking, the input → symptom → fault-candidate table (ML classification), residual-based fault evidence.
Absorbs	Health › Deductions as the Points lost column.
Drops	Log (→ Trends › Events), validity strip (→ Sensors), Actions (→ Maintenance; each rule row keeps a link to its action).

`01` **Alarms**

`Columns: Pri. · Alarm · State · Persist. · Clear · In alarm`

- Performance over 1 h / 8 h / 12 h; lifecycle pending → active → clearing

`02` **Rules**

`Columns: Rule · Lane · Persist. / clear · Points lost · Last fired · Logic`

- All 21 rules including thermal R-SHL-* and data-quality R-DQ-*; segmented sample counters; clicking a rule scopes the page

`03` **Evidence**

`Columns: Sensor · Value · Limits · vs alert / alarm · Residual 1 h · Cited by`

- Only sensors a firing rule reads. Residual shown as a fact; its interpretation is Analysis
- Vibration severity: Point · Zone · Zone limits (mm/s)

Thermal

`kiln only` `thermal-view.tsx`

Question. Is the kiln shell thermally safe, and is the refractory or coating changing?

Shell max | Above model | Hottest section | Scanner coverage

Shell Zones Hot spots `STICKY`

Shell

Kiln over scanner map to scale





Shell map · metres × angle

Axial profile

Scanner · coverage · behind tyres · last map · → Sensors

Zones

Per zone · lining estimate

Hot spots

Ranked

Circumferential



Section trend

Drops Rules section (→ Diagnosis › Rules, Thermal filter) and the separate Scanner card (one panel under Shell).

---

01 Shell

- Scanner map under the kiln drawing; axial profile on the same metre axis
- Scanner panel: coverage, obstructions, last map, link to the sensor in Sensors

---

02 Zones

Columns: Zone · Max · Mean · Model · Deviation · Area ≥ alert · Lining / coating · Status

---

03 Hot spots

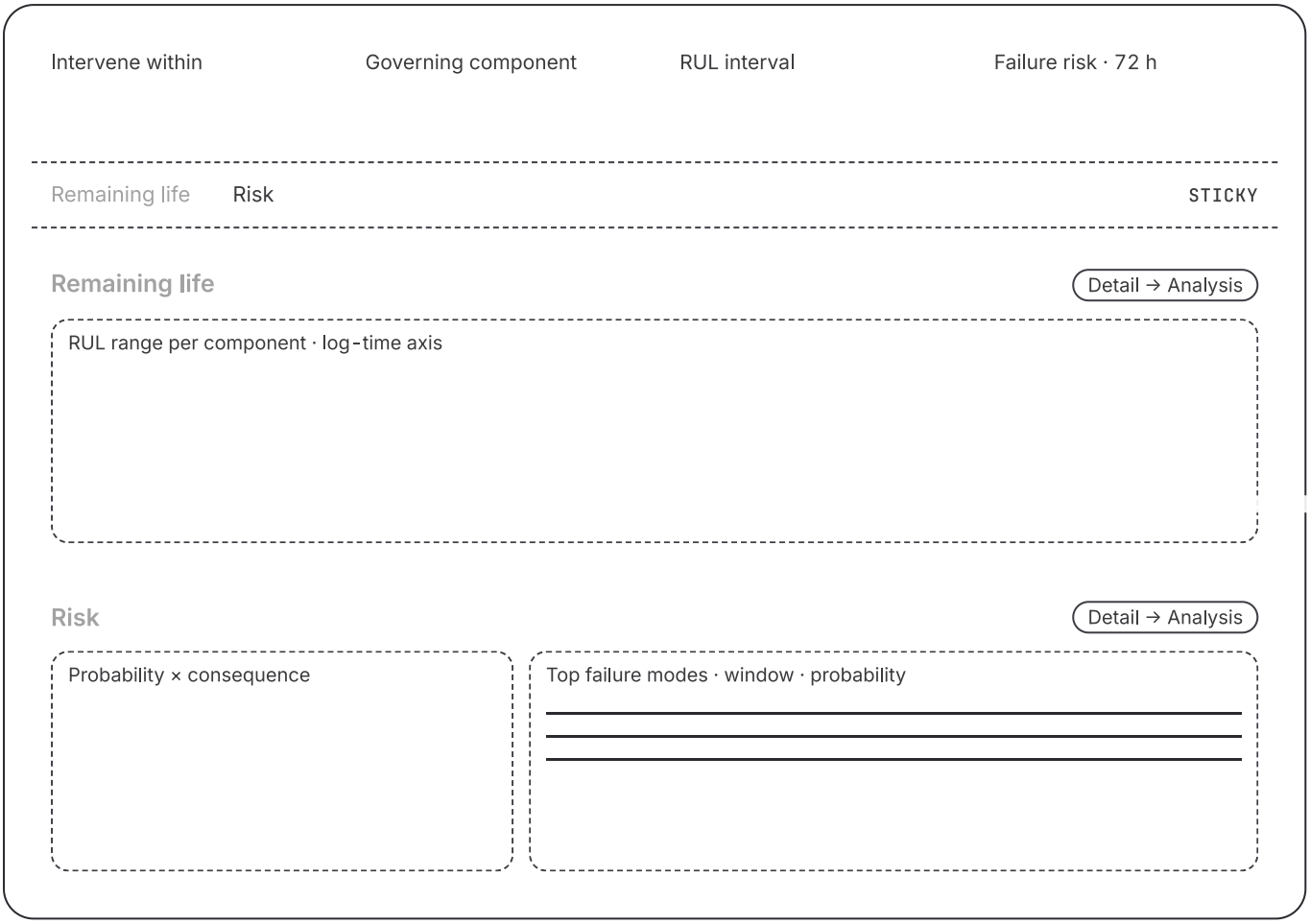
Columns: # · Position · Max · At · Rate

- Selected row drives the circumferential plot and section trend

- Selected row drives the circumferential plot and section trend

Prognosis `prognosis/prognosis-page.tsx`

Question. What happens next, and when do I need to act? (the brief)



Role	A one-screen brief of the prediction capabilities (Analysis 5–8). Each card shows the current figure and a link; method, intervals, assumptions and limitations are in Analysis.
Moved to Analysis	Signal forecasts (cones), health projection and P-F bracket, degradation evidence, confidence, accuracy.
Hands off	Windows and recommended timing live on Maintenance.

---

01 KPI strip

- Intervene within · Governing component · RUL interval · Limit crossings in 24 h

---

02 Remaining life

`Columns: Component · RUL range · Status`

- One bar per component; row link opens Analysis › Remaining life for that component

---

03 Risk

`Columns: Failure mode · Probability · Window · Consequence`

- Matrix plus the top modes; no evidence here

Analysis

longest page · 11 sections

analysis/ · one section per AI capability

Question. What does the AI see, why does it say so, and how far ahead can it be trusted? One section per capability in the three specifications, in pipeline order.

Health score | Anomaly score | Top fault · confiden... | Degradation indica... | Failure risk · horizon

RUL · P50

Health Expected Anomaly Fault Degradation Forecast Risk RUL Explain Model Feedback STICKY

1 · Health & condition

Machine | Components | Process

Health score · machine → component → process · Δ over time

Measurements contributing to reduced health

2 · Expected behaviour

All | Condition | Process

Actual vs expected · deviation · uncertainty band · affected equipment

Deviation matrix · signal × time · |σ|

3 · Unknown anomaly

Anomaly score over time · detection time

Unusual relationships · sensor × sensor

Affected measurements · temporal anomalies · known-fault match: none

- [ ]
- [ ]
- [ ]

4 · Issue source & fault

Issue location on the flow (PFD / P&ID) · affected section

Sensor → Card → Gateway → Broker → ULTRON

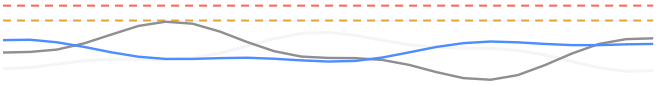
Fault ranking · calibrated probability

Diagnostic evidence · alternative hypotheses · verification required

- [ ]
- [ ]
- [ ]

5 · Degradation

Learned degradation indicator over time · per component



Stage 0–5 per component

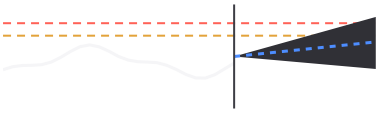
Contributing parameters · assessment limitations

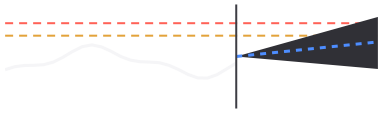
- [ ]
- [ ]
- [ ]

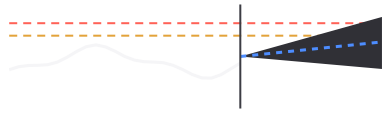
6 · Future condition

24 h 7 d

Forecast + prediction interval







Target · horizon · trend · operating assumptions · interpretation

- [ ]
- [ ]
- [ ]

7 · Failure risk

72 h 30 d

Risk trend · probability as observations update

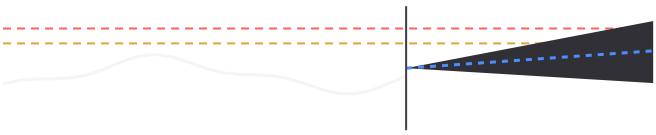
Probability × consequence

Failure event · horizon · probability · contributing factors · reliability / unavailable reason

8 · Remaining useful life

Component

RUL trend · P10 / P50 / P90 · end-of-life criterion



Target · EoL definition · usage assumptions · evidenc...

9 · Explainability

Prediction

Feature importance · contribution breakdown (SHAP)

Plain-language explanation · model evidence · limit...

Model quality

Precision · Mean lead · Brier · Inputs valid · Model versions

Calibration



Model card per capability · excluded inputs · pipeline status

Feedback

Open findings · Confirm fault / False alarm / New normal

Scope	Every AI / ML / DL capability from the three specifications: four monitoring, four prediction, plus explainability. The fifth prediction capability (maintenance recommendation) is shown on the Maintenance tab with a link back to its supporting predictions here.
Section pattern	Each capability card shows its Expected Output fields, the operating context it applies to, and an explicit limitation / unavailable line when evidence is insufficient (every spec requires it).

Entry points	Prognosis cards link to 5–8; Maintenance actions link to the predictions behind them; Overview health links to 1; the machine bar lead finding links to 4 or 3; each finding row offers Explain (10).
Order	Pipeline order: is it healthy → expected vs actual → unfamiliar pattern → where / what fault → is it worsening → what next → how likely → how long → why.

---

01 **KPI strip**

- Health score · Anomaly score · Top fault with confidence · Degradation indicator · Failure risk with horizon · RUL P50

---

02 **1 · Health & condition assessment**

`Columns: Asset · Health · Δ · Abnormal · Contributing measurements`

- Machine, component and process health scores against the healthy reference; change over time
- Measurements contributing to reduced health
- Spec: Process & Condition Monitoring §1

---

03 **2 · Expected behaviour prediction**

`Columns: Signal · Expected · Actual · Deviation · Interval · Affected`

- Expected values for the current operating context (regression / time-series model) vs measured
- Significant deviations flagged; deviation matrix over time
- Spec: §2

---

04 **3 · Unknown anomaly detection**

`Columns: Measurement · Contribution · Sequence · Since`

- Anomaly score over time with detection time and development
- Unusual sensor relationships and timing; explicit 'no known-fault match' status
- Spec: §3

---

05 **4 · Issue-source identification & fault diagnosis**

`Columns: Hypothesis · Location · Probability · Evidence · Verify by`

- Likely affected section on the process flow; most likely supported fault type
- Ranked alternatives with calibrated confidence only where validated; unknown-fault status
- Inspections or measurements required to confirm
- Spec: §4

---

06 **5 · Degradation assessment**

`Columns: Component · Indicator · Stage · Progression · Limitation`

- Learned degradation indicator on a validated 0–5 scale, per component, over time
- Contributing parameters; whether evidence supports a reliable interpretation
- Spec: Prediction §1

---

07 **6 · Future condition prediction**

`Columns: Target · Horizon · Predicted · Interval · Trend · Assumptions`

- Forecasts with prediction intervals for the selected parameters and horizon (24 h, 7 d)
- Operating assumptions stated; interpretation of the predicted change
- Spec: Prediction §2

---

08 7 · Failure risk prediction

Columns: Failure event · Horizon · Probability · Trend · Factors · Reliability

- Probability of the defined failure within the horizon; risk trend as observations update
- Contributing factors; calibration and applicability; explicit reason when unavailable
- Spec: Prediction §3

---

09 8 · Remaining useful life

Columns: Component · RUL · Interval · EoL definition · Trend · Assumptions

- Hours or cycles to the defined end-of-life event with prediction interval
- RUL trend as measurements arrive; whether lifetime evidence is sufficient
- Spec: Prediction §4

---

10 9 · Explainability

Columns: Feature · Contribution · Direction · Reference

- Pick any prediction above; feature importance and contribution breakdown (SHAP where supported)
- Plain-language explanation, model evidence (inputs, reference, version), and the model-influence vs physical-cause distinction
- Spec: Explainable AI

---

11 Model quality

- Calibration, precision, mean lead, Brier; model card and version per capability; inputs excluded by validation; pipeline stage status

---

12 Feedback

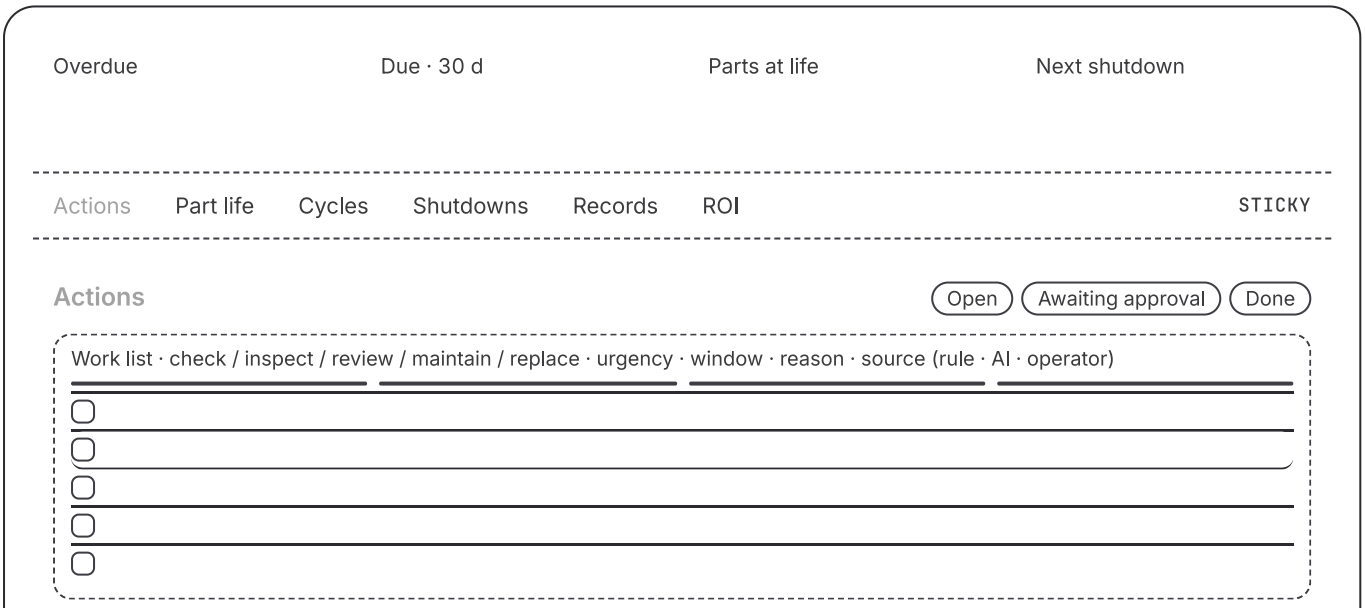
Columns: Finding · Since · Label · By

- Operator labels per open finding; they feed the ledger and retraining

Maintenance

new · from MOM Sensor Intelligence brief

Question. What work does this machine need, which parts are near end of life, when is the next shutdown, and what is the monitoring worth?



Selected action: what goes wrong if it continues · supporting evidence → Diagnosis / Analysis · approve · create work order

P7 · TMP-04 312 °C Trends Diagnosis

Part life

Wear parts All

Part · life units · installed · usage · remaining · status ORANGE / YELLOW / GREEN

☐			
☐			
☐			
☐			
☐			

Controls

Inspection does not reset life | Continued use needs approval | Status never overrides an active fault | Expected life ≠ failure date

Cycles

Interval timeline · last done · next due · overdue

Inspection findings · replacement history

☐			
☐			
☐			

Shutdowns

Shutdown schedule · affected equipment · work duration

Replacement timing · within N days · reason · linked shutdown · evidence basis

☐			
☐			
☐			

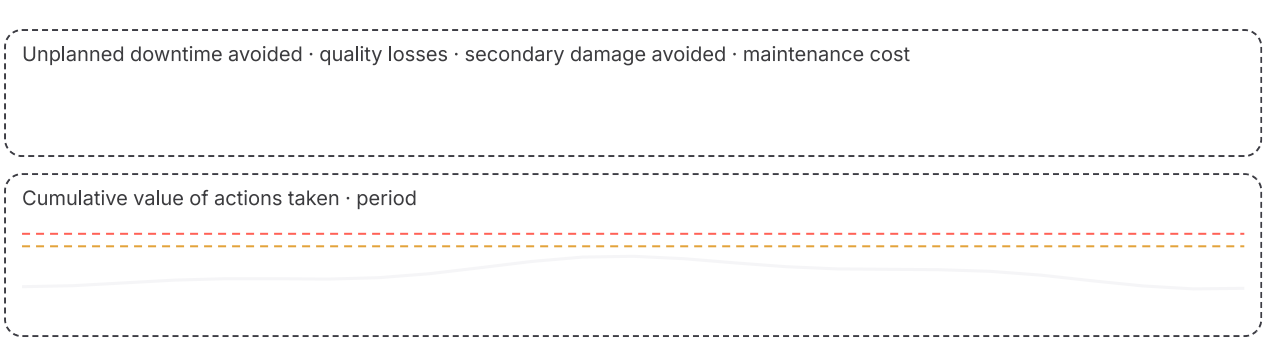
Records

Survey Maintenance Upload

Record · machine / part · source · activity date · uploader · evidence

☐			
☐			
☐			
☐			

ROI



Source	MOM §1 (recommend action), §2 Part life & maintenance tracking, §3 Survey & maintenance data reliability, §4 Shutdown & replacement planning, §5 ROI.
Takes from Diagnosis	Actions (rule-library prescriptive actions).
Takes from Analysis	§9 Predictive maintenance recommendation; Analysis keeps the supporting predictions and links here.
Takes from Prognosis	Next window.
Rules	Inspection alone never resets part life. Continued use past entered life needs an acceptable inspection and approval. Part-life colour never overrides an active fault. A replacement date is suggested only when condition evidence, part life, inspection findings and approved rules all support it.

---

01 KPI strip

- Overdue work · Due in 30 d · Parts at entered life (orange) · Next shutdown with available duration

---

02 Actions

`Columns: Action · Kind · Urgency · Window · Reason · Consequence if ignored · Source · Status`

- Kinds: measurement check, inspection, process review, maintenance, replacement
- Source is a rule (Diagnosis), an AI recommendation (Analysis 9, always 'requires engineering review') or an operator
- Approve → work order; feeds Home › Ops center

---

03 Part life

`Columns: Part · Life units · Installed / replaced · Accumulated usage · Remaining · Status · Next deadline`

- ORANGE: entered life reached, thorough inspection required
- YELLOW: inspected, not replaced; findings and next review / replacement deadline recorded
- GREEN: replaced; new life cycle started

---

04 Cycles

`Columns: Task · Interval · Last completed · Next due (date / usage) · Overdue · Last findings`

- Replacement history per part

---

05 Shutdowns

`Columns: Shutdown · Date · Equipment · Duration · Planned work`

- Replacement timing suggestions state the reason and link to a suitable shutdown

---

06 Records

`Columns: Record · Machine / part · Source · Activity date · Uploader · Evidence`

- Survey and maintenance uploads linked to the correct machine or part; missing fields block the upload

---

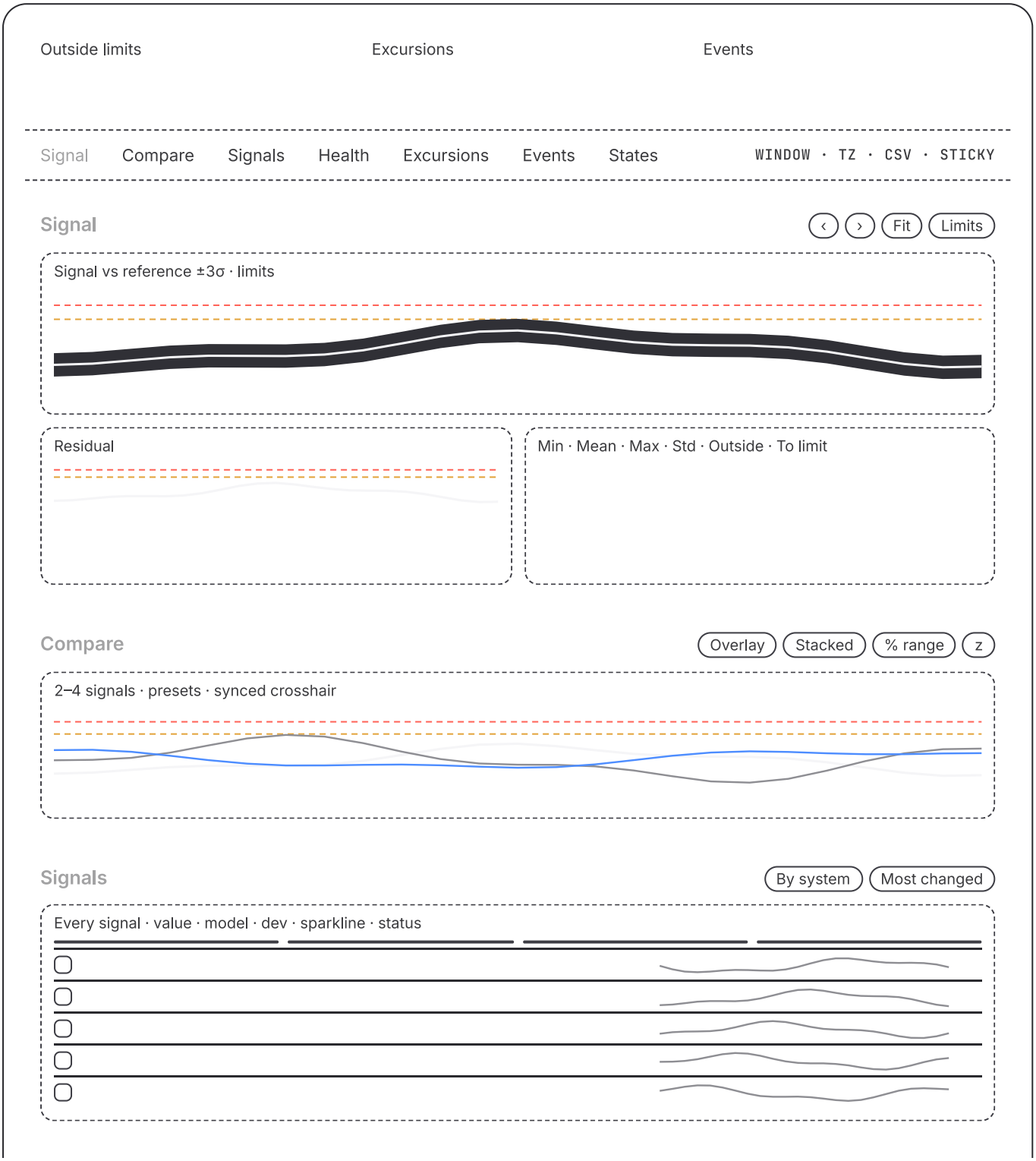
`07` ROI

`Columns: Measure · Period · Value · Basis`

- Reduced unplanned downtime, lower quality losses, avoided secondary damage, maintenance-cost improvement; each tied to the actions that earned it

Trends `trends-view.tsx`

Question. How have signals, health and events evolved over time?



Health

Machine + system health · 85 / 60 · rule events (was Health › Trend)



Excursions

Periods outside limits or model band

☐			
☐			
☐			

Events

All · Alarms · Operator · Data quality · Scenario

The one event log

☐			
☐			
☐			
☐			

States

On / off timelines

Absorbs	Health › Trend; every other page's event log.
Window controls	Sit in the section index row, not in a header.

---

01 Signal

- One signal vs reference model ±3σ and limits, residual panel, window stats; step prev / next

---

02 Compare

- Overlay or stacked, % range or z-score; D5 9.5 relationship presets

---

03 Signals

`Columns: Signal · Latest · Model · Dev. · Min · Max · sparkline · Status`

- The full signal table. Overview, Sensors and Diagnosis link here

---

04 Health

- Machine and system health history with 85 / 60 limits and rule markers

---

05 Excursions

`Columns: Signal · Band · Start · Duration · Peak · Rule`

06 Events

Columns: Time · Type · Event · Source

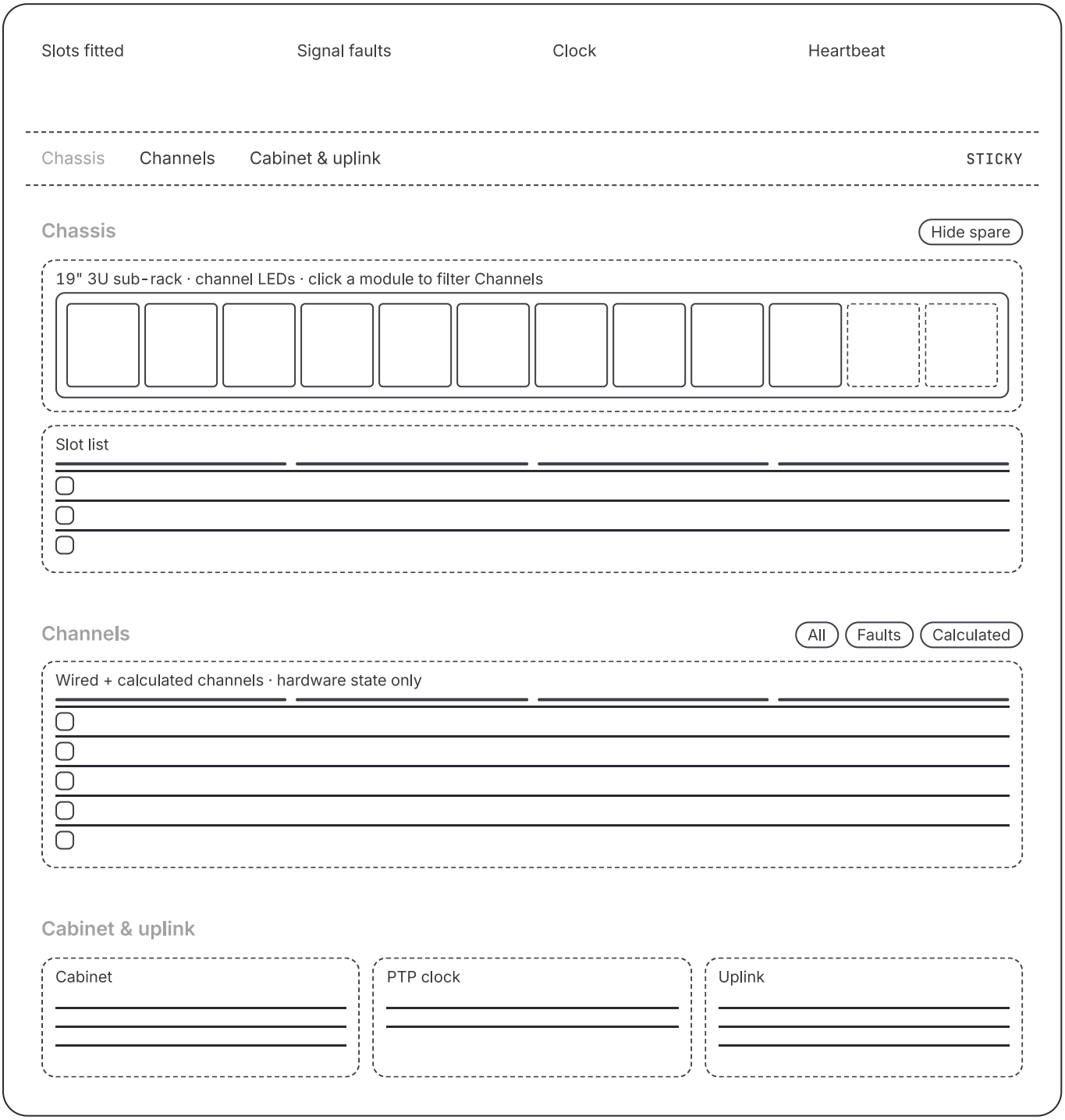
- Engine, operator and scenario events; filter presets used by Diagnosis and Simulation links

07 States

Columns: State · Off-normal · Changes

Rack `rack-view.tsx`

Question. How is the acquisition hardware set up, and is it healthy?



Merges Slots into Chassis (drawing + list); Calculated into Channels as a filter.

Drops Process-state column (Trends) and the Sensors KPI (register is in Sensors).

01 Chassis

`Columns: Slot · Module · Type · Signal · Channels · State`

- Drawn sub-rack; LEDs dark only for invalid / suspect channels

02 Channels

`Columns: Location · Sensor · Register · Sampling · Reading · Device state`

- Calculated values listed with Inputs · Basis under the same filter

03 Cabinet & uplink

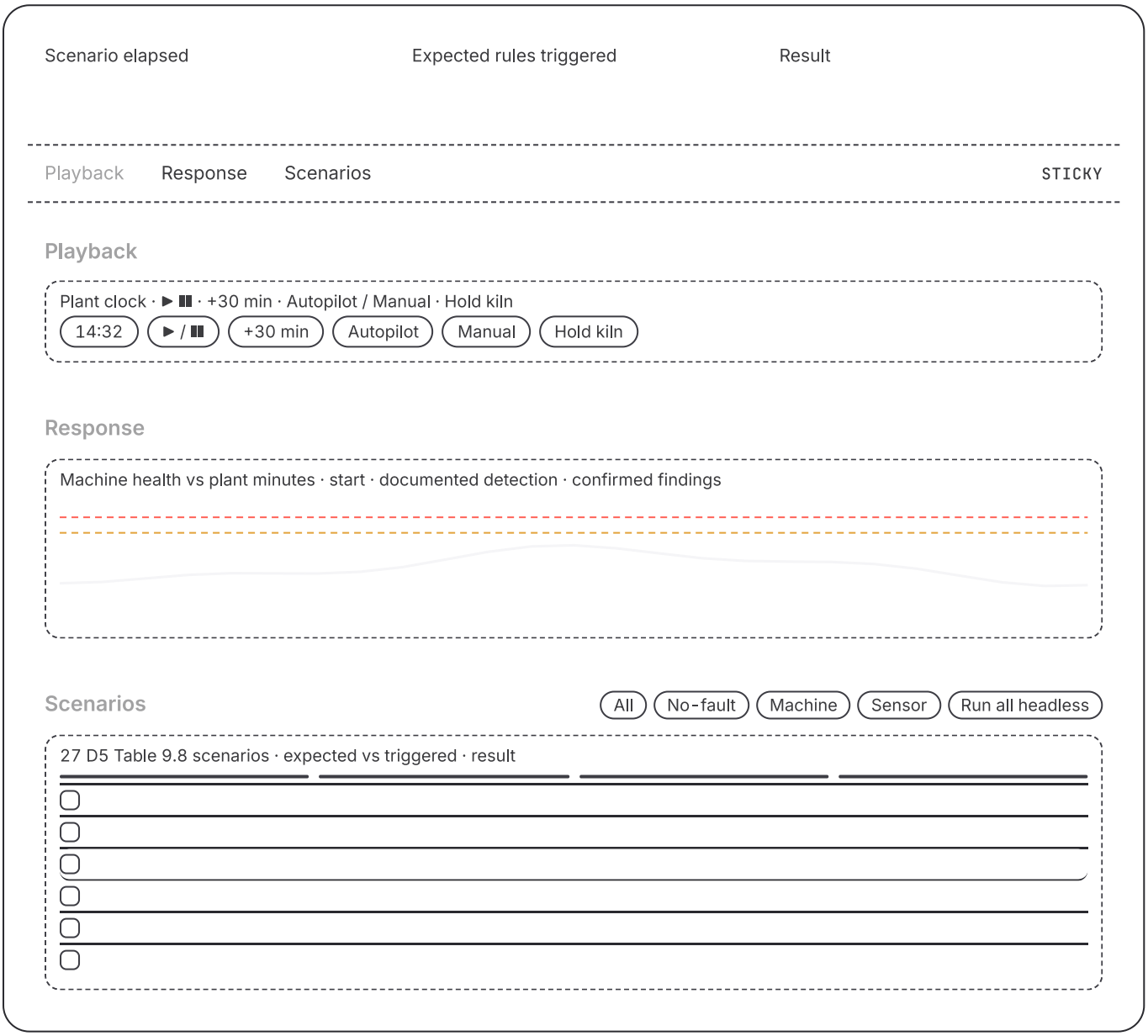
`Columns: Item · Recorded · Specified`

- Enclosure, firmware, power rails; PTP lock and path delay; gateway, broker, packet loss, buffering

Simulation

`kiln only` `simulation-view.tsx`

Question. Which scenario is running, and did the plant respond as documented?



Drops Findings raised (→ Diagnosis › Alarms) and Event log (→ Trends › Events, Scenario filter).

---

01 **Playback**

- Shared kiln-sim fleet controls; healthy baseline; the headless run uses its own engine

---

02 **Response**

- Live health across the scenario with the documented detection time and each confirmed finding

---

03 **Scenarios**

`Columns: Scenario · Expected · Triggered · Result`

- Pass / Fail / Not run; outcome labels (Quiet as expected, False alarm, Developing, Not yet detected)

Proposed structure derived from `src/components/assets/**`. Directory page unchanged. Devices, AssetIQ and Configure untouched; AssetIQ › Accuracy remains the fleet-level view of the same ledger.